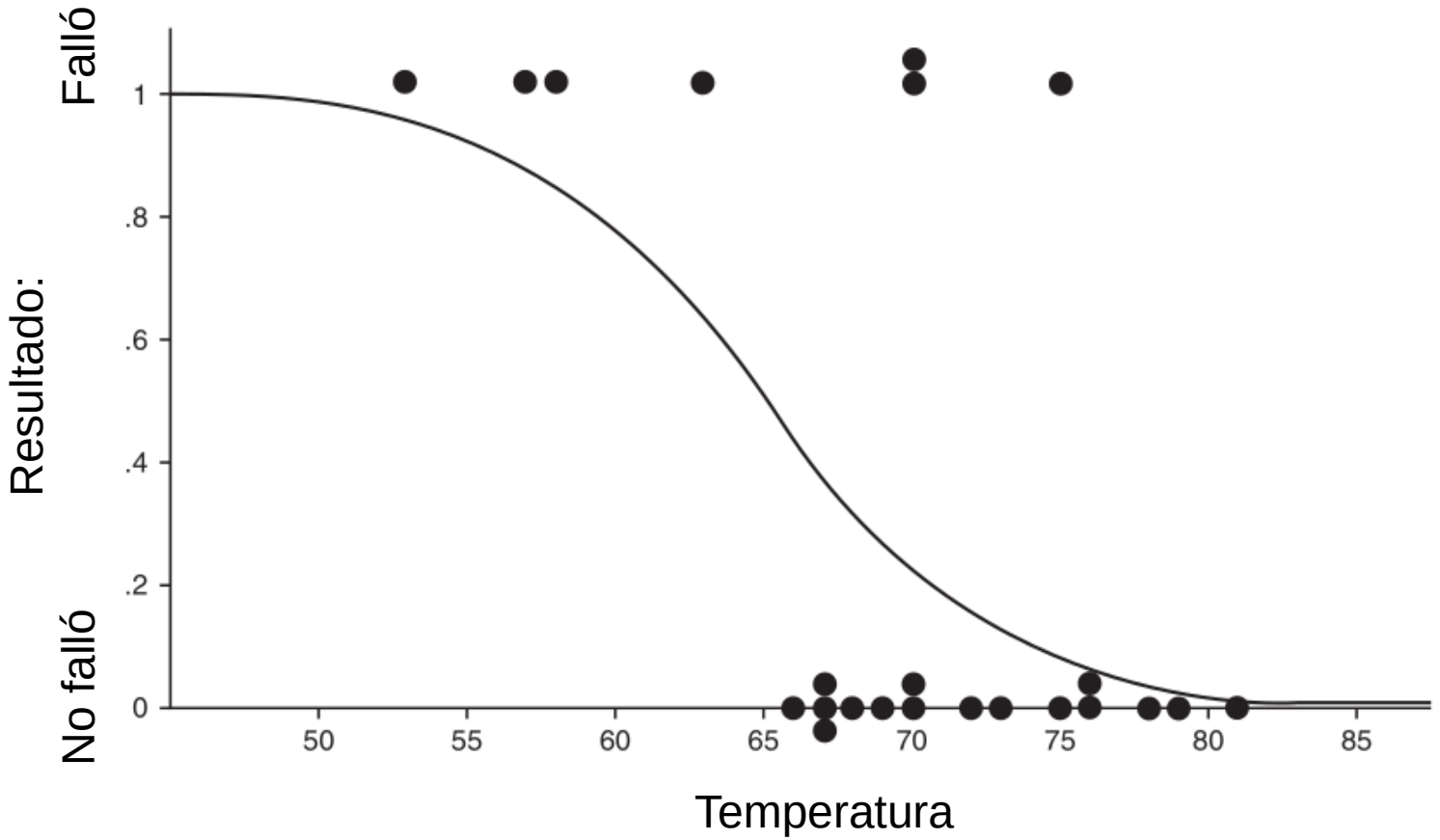
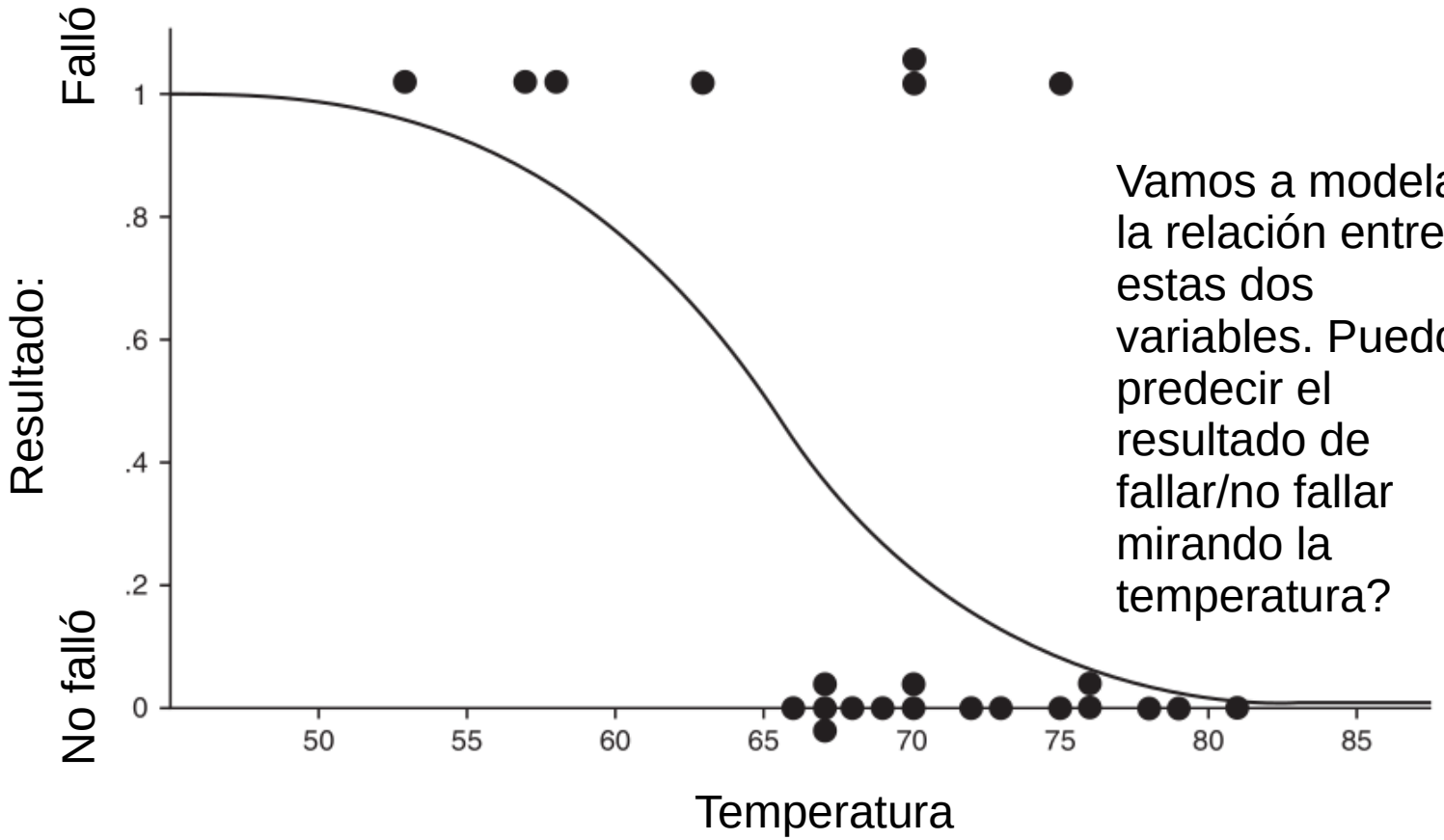
Regresión logística

(regresión: relación entre dos variables)



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(regresión: relación entre dos variables)

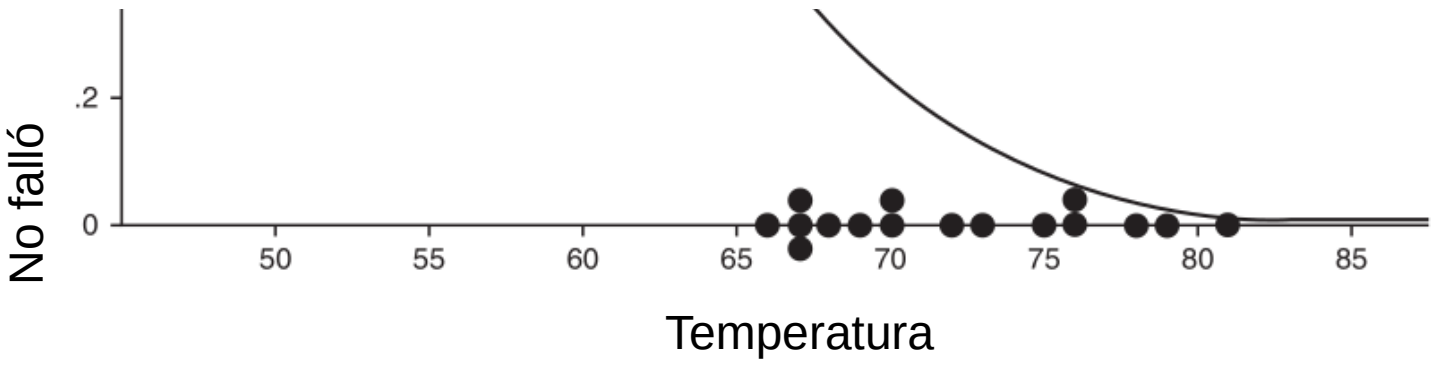


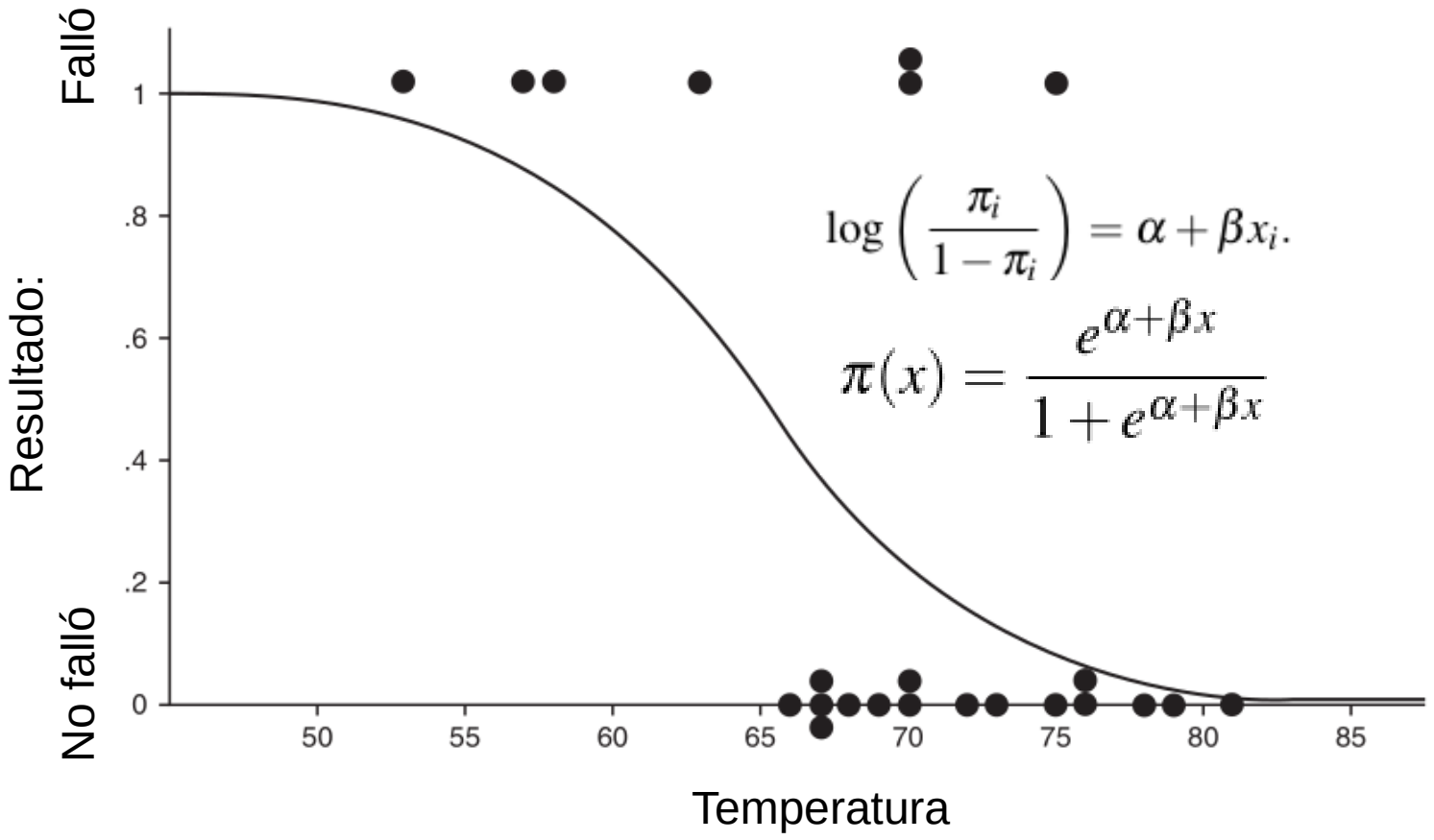
Vamos a modelar la relación entre estas dos variables. Puedo predecir el resultado de fallar/no fallar mirando la temperatura?



Another very useful GLM is the *logistic regression model*. In this model, the responses $Y_1, \ldots, Y_n$ are independent and $Y_i \sim \text{Bernoulli}(\pi_i)$. (The Bernoulli family is an exponential family.) Recall, $\text{E}Y_i = \pi_i = P(Y_i = 1)$. In this model, π_i is assumed to be related to x_i by

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \alpha + \beta x_i. \qquad (12.3.1)$$

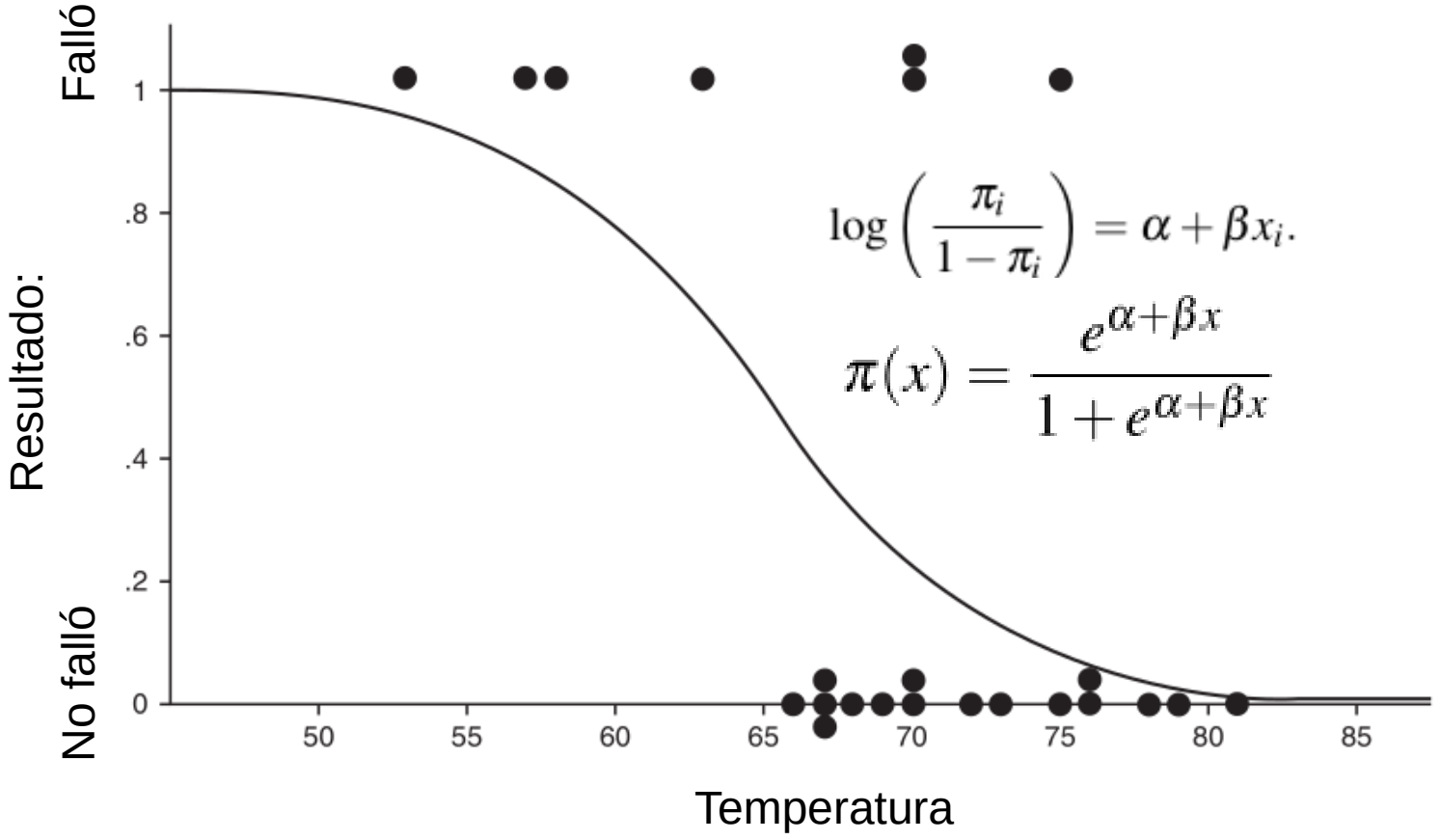



Resultado:
Falló
No falló
1
.8
.6
.4
.2
0
50
55
60
65
70
75
80
85
Temperatura
$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \alpha + \beta x_i.$
$\pi(x) = \frac{e^{\alpha+\beta x}}{1+e^{\alpha+\beta x}}$

Interpretación de los parámetros:

$$\log\left(\frac{\pi(x+1)}{1-\pi(x+1)}\right) - \log\left(\frac{\pi(x)}{1-\pi(x)}\right) = \alpha+\beta(x+1)-\alpha-\beta(x) = \beta.$$

Alfa es el log-odds de éxito cuando x=0.
Beta es el cambio en el log-odds de éxito correspondiente a un incremento de 1 unidad en x
(en regresión lineal era el cambio en el valor medio de Y correspondiente a un incremento de 1 unidad en x)



Contraste de hipótesis para beta

$$\lambda(\mathbf{x}) = \frac{L(\hat{\theta}_0|\mathbf{x})}{L(\hat{\theta}|\mathbf{x})}$$

Likelihood ratio test (LRT)

It is, perhaps, most common, in this model, to test the hypothesis $H_0 : \beta = 0$, because, as in simple linear regression, this hypothesis states there is no relationship between the predictor and response variable. The Wald test statistic, $Z = \hat{\beta}/\text{se}(\hat{\beta})$ has approximately a standard normal distribution if H_0 is true and the sample size is large. Thus, H_0 can be rejected if $|Z| \geq z_{\alpha/2}$. Alternatively, H_0 can be tested with the log LRT statistic

$$-2\log\lambda(\mathbf{y}^*) = 2[\log L(\hat{\alpha}, \hat{\beta}|\mathbf{y}^*) - L(\hat{\alpha}_0, 0|\mathbf{y}^*)],$$

where $\hat{\alpha}_0$ is the MLE of α assuming $\beta = 0$. Using standard binomial arguments (Exercise 12.20), it can be shown that $\hat{\alpha}_0 = \sum_{i=1}^{n} y_i/n = \sum_{j=1}^{J} y_j^*/\sum_{j=1}^{J} n_j$. Therefore, under H_0, $-2\log\lambda$ has an approximate χ_1^2 distribution, and we can reject H_0 at level α if $-2\log\lambda \geq \chi_{1,\alpha}^2$.

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